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Peer Review File

A Transparency Statement Improves Trust in Community-Police Interactions

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This manuscript has been previously reviewed at another journal. This document only contains information relating to versions considered at Nature Communications.

Parts of this Peer Review File have been redacted as indicated to remove third-party material.

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Attachments originally included by the reviewers as part of their assessment can be found at the end of this file.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
I previously reviewed this paper. I appreciate the additional work on voice prosody and further description of the campus setting. I think the skin conductance findings show general arousal/intensity rather than anything more specific.

The authors have done an enormous amount of work on a fascinating topic. This is a strong paper that should be published.

Reviewer #2

(Remarks to the Author)
This study has serious shortcomings, which together mean that it adds little to the literature. Essentially, it uses a worse intervention and worse samples that past experiments, and just shows what we already know—namely, that exposure to procedural justice (here called “transparency statements”) has a short-term, encounter-specific effect on attitudes toward the particular officer(s) in the interaction. To be crystal clear, the biggest limitations of the study are that it uses a weak measure of procedural justice (see below), calls it something different (see below), and then focuses only on specific (rather than general) policing attitudes. Let me explain further.

First, the authors have pulled out a single component (i.e., demonstration of trustworthy motives) from the broader procedural-justice package that has been tested repeatedly in much stronger field trials, after which they acted like it was something new, calling it “transparent communication of benevolent intent,” without even seriously (or thoroughly) discussing the other trials. The literature is unequivocal that the linguistic demonstration of “trustworthy motives” is a central component of procedural justice (Mazerolle et al., 2013, p. 35, 41). Now, let me point to just a few of the trials that have tested procedural justice, only one of which is even cited (on p. 4). Mazerolle et al. (2013) did a field experiment (N = 2,746) randomizing the language officers used in actual traffic stops, a component of which was telling the civilians who were stopped that they were not singled out and that the stops were important to save lives (i.e., a transparent statement of benevolent intent). Similar field experiments were conducted by many others, such as Sahin et al. (2017), MacQueen and Bradford (2015), Terpstra et al., (2023), and Weisburd et al. (2022). And none of these experiments were conducted in the kind of artificial conditions used in the current study (e.g., telling participants they were in a study about natural interpersonal interactions, outfitting them with physiological sensors, and then having them have an odd conversation with an officer). That is, all prior field trials used actual police-civilian interactions with potential enforcement consequences (e.g., ticket, arrest).

Second, beyond the field trials discussed above, there is a huge body of survey experiments that basically do the same thing

(but with better interventions and better outcomes) than the current study does in experiments 2 through 7 (e.g., Johnson et al., 2017; Lowrey et al., 2016; Maguire et al., 2016; Roche, 2019). That is, the last six experiments in the current study look very similar to those that appear in mid-level special journals, and really add nothing new over the findings in those mid-tier articles.

Third, and this is very important, the key finding from past experimental work—both from the field trails and survey experiments—is that procedural justice interventions, like transparency statements demonstrating trustworthy motives (or benevolent intent), mainly affect encounter-specific attitudes, but have little effect on general policing attitudes (e.g., Johnson et al., 2017; Lowrey et al., 2016; Maguire et al., 2016; Sahin et al., 2017). Nagin and Telep (2016, p. 10) emphasize this point: "There was a significant impact only on these encounter-specific outcomes and not on more general views about the police." This is a huge problem for the current study where the entire focus is on encounter-specific attitudes and little attention is given to general policing attitudes. That is, in the current study the key outcomes are perceptions of the specific officer in the interaction (e.g., p. 10: "The officer just wanted to make friendly conversation"). The outcomes are NOT general policing attitudes. This is true even in the later experiments where some questions ask about the specific officer or "another officer from this organization" (e.g., how likely to "wave if you see this officer", p. 30). So, what the study basically shows is that if an officer explicitly says they just want to make friendly conservation (e.g., "I'm just out and about walking around, talking to people in the community"), then civilians are more likely to believe that the specific officer is just trying to have a friendly conversation. Although this is interesting, it is not groundbreaking.

I would also emphasize that on page 4, the current authors are misrepresenting the literature when they claim that the results of past work are mixed. They cite Bolger and Waters (2019), but that meta-analysis concluded that there was strong support for the theory of procedural justice: "The analysis produced results in support of all three causal chains of Tyler's theory … Citizen perceptions of police procedural justice would appear to have both a direct impact and indirect effect on citizen cooperation, the latter by way of police legitimacy beliefs" (p. 98). That meta-analysis also was not focused on experiments or field trials, but mostly included cross-sectional surveys looking at associations between perceptions and beliefs; so it is not really relevant here, where the specific focus is on experimental effects of interventions, both in the current study and in the relevant past work (e.g., Johnson et al., 2017; Lowrey et al., 2016; MacQueen & Bradford, 2015; Maguire et al., 2016; Mazrolle et al., 2013; Sahin et al., 2017; Weisburd et al., 2022). What a review of all the experiments shows is that they only meaningfully affect encounter-specific perceptions, just like the current authors find.

Shockingly, the authors say on pages 15-16 that "people of color and those from lower-SES contexts might have more negative attitudes toward law enforcement" and then they cite a Gallup poll. This is not a "might have" issue. There is a tremendous amount of very strong research that shows quite clearly that such Americans do have more negative policing attitudes and emotions (e.g., Pickett et al., 2022).

Finally, I would also emphasize that whereas the authors seemed surprised that they found "no consistent moderation by demographic group" (p. 16), they should not be. Prior research shows that procedural justice has similar effects across demographic groups (Wolfe et al., 2016). And, again, all the current authors are doing is testing the effect of a single dimension of procedural justice, what Mazerole et al. (2013, p. 35, 41) call demonstrating of "trustworthy motives."

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Reviewer #3

(Remarks to the Author)
Please see the attached PDF document.

Version 1:

Reviewer comments:

Reviewer #3

(Remarks to the Author)
Please see the attached file.

Reviewer #4

(Remarks to the Author)
Thank you for the opportunity to review the manuscript, A Transparency Statement Improves Community-Police Interactions. The authors triangulate across qualitative, field experimental, and web experimental methods to test the claim that having officers transparently state their intentions when contacting the public has salutary effects on these interactions. This work is timely, theoretically rich, empirically rigorous, and practically useful. The authors advance several fields, and this will be an impactful contribution.

As a substitute reviewer, I was tasked with evaluating the revised manuscript in light of the previous reviewer's comments and the authors revisions. To cut to the chase, I believe that this work is of sufficient rigor and novelty for publication, and the reviewers have responded to the previous reviewer's comments satisfactorily, with one exception. As I describe below, the authors need to provide more clarity about the relationship and novelty of the work vis a vis procedural justice. While I believe this change is important for understanding the impact of the work, I would consider it a minor revision. Here, I address the reviewer's comments -and authors' response- one by one.

1. First, the authors have pulled out a single component (i.e., demonstration of trustworthy motives) from the broader procedural-justice package that has been tested repeatedly in much stronger field trials, after which they acted like it was something new, calling it "transparent communication of benevolent intent," without even seriously (or thoroughly) discussing the other trials.

This critique has two parts: first, the novelty of the contribution vis a vis other research on procedural justice: whether the authors are claiming discovery of a feature of PJ that other interventions have tested. Second, these data are simply restating past demonstrations of procedural justice.

On the second point, I think the authors provide a compelling response detailing how this examination differs from other examinations of procedural justice. The field experiment here is weighted more towards internal versus external validity of the field experiments by Mazerolle and Telep; however, they capture an extremely rare outcome: citizens' perceptions of the actual interaction intervened on. Likewise, procedural justice is often trained as an orientation, with suggested behaviors; this more precise manipulation of one of those behaviors provides a precise test of one of those behaviors. So, the research is not an inferior version of past experiments, nor is it a retread.

On the first point, I agree with the reviewer in principle (if not vehemence). Communicating trustworthy motives is a core feature of procedural justice, which has been the dominant model of police-citizen interactions for the past thirty years. The contribution here is not a retread, but there is a close conceptual relationship: I would characterize it as unpacking how these motives are communicated -or not- in interactions. That is a contribution (as they point out, the operationalizations of PJ are many and often deployed as a package or with maximal officer discretion), but the introduction as written treats PJ as a parallel approach that is incidentally related to their intervention.

The authors respond that transparency statements are parallel, in that their insight came about from qualitative observation. I do not find this convincing. Ride-along observations reveal a lot of things that research has examined: the investigatory nature of many stops, police officers' construals of communities, and so on. It would be fallacious to claim to discover any of these, as it is that the authors have discovered that officers' motives are often opaque to community members.

I don't think the authors actually believe they've discovered transparency, but they have identified an element of

communication that is central but unstudied in how transparency is or isn't communicated. I believe the introduction should more clearly identify this contribution and its relation to the voluminous research on procedural justice, earlier rather than later. One simple way to do so is to identify procedural justice as a popular means through which the ideals of trust and legitimacy are pursued, the ambiguities of what PJ looks like in practice, and then their astute observations of how the trustworthy ideals of community policing are lost when officers are not transparent.

2. Second, beyond the field trials discussed above, there is a huge body of survey experiments that basically do the same thing (but with better interventions and better outcomes) than the current study does in experiments 2 through 7 (e.g., Johnson et al., 2017; Lowrey et al., 2016; Maguire et al., 2016; Roche, 2019). That is, the last six experiments in the current study look very similar to those that appear in mid-level special journals, and really add nothing new over the findings in those mid-tier articles.

This objection is, as the authors point out, quite vague. What these experiments add is an unpacking of the process believed to underlie the authors' field experiment. The studies the reviewer cites lack the field experimental component of this package. The triangulation of designs is a strength of this paper.

3. Third, and this is very important, the key finding from past experimental work—both from the field trials and survey experiments—is that procedural justice interventions, like transparency statements demonstrating trustworthy motives (or benevolent intent), mainly affect encounter-specific attitudes, but have little effect on general policing attitudes (e.g., Johnson et al., 2017; Lowrey et al., 2016; Maguire et al., 2016; Sahin et al., 2017). Nagin and Telep (2016, p. 10) emphasize this point: "There was a significant impact only on these encounter-specific outcomes and not on more general views about the police."

This is a huge problem for the current study where the entire focus is on encounter-specific attitudes and little attention is given to general policing attitudes. That is, in the current study the key outcomes are perceptions of the specific officer in the interaction (e.g., p. 10: "The officer just wanted to make friendly conversation"). The outcomes are NOT general policing attitudes. This is true even in the later experiments where some questions ask about the specific officer or "another officer from this organization" (e.g., how likely to "wave if you see this officer", p. 30). So, what the study basically shows is that if an officer explicitly says they just want to make friendly conservation (e.g., "I'm just out and about walking around, talking to people in the community"), then civilians are more likely to 7 believe that the specific officer is just trying to have a friendly conversation. Although this is interesting, it is not groundbreaking.

As the authors address, the procedural justice literature is mixed as to whether and when elements of specific encounters generalize to views of organizations and institutions. I also agree with them that the intent behind their study is to understand what features of conversations lead to these impressions of trustworthiness. The focus on interactions is not a fatal flaw for their research question. The linguistic components of interactions which lead to these attributions are not well known and so this constitutes an advance of our knowledge of procedural justice.

4. I would also emphasize that on page 4, the current authors are misrepresenting the literature when they claim that the results of past work are mixed. They cite Bolger and Waters (2019), but that meta-analysis concluded that there was strong support for the theory of procedural justice: "The analysis produced results in support of all three causal chains of Tyler's theory … Citizen perceptions of police procedural justice would appear to have both a direct impact and indirect effect on citizen cooperation, the latter by way of police legitimacy beliefs" (p. 98)…

The authors have addressed this point satisfactorily in their systematic review of the treatment effects of procedural justice.

5. Shockingly, the authors say on pages 15-16 that "people of color and those from lowerSES contexts might have more negative attitudes toward law enforcement" and then they cite a Gallup poll. This is not a "might have" issue. There is a tremendous amount of very strong research that shows quite clearly that such Americans do have more negative policing attitudes and emotions (e.g., Pickett et al., 2022).

The authors address this point, although I would add that the Pickett study they cite (in response to the review comment) in discussing the magnitude of racial gaps in perceptions of the police is slightly misleading. Fear of being killed is a quite an extreme outcome, and other studies have certainly documented large and durable gaps in trust (two examples below). The authors may find them useful in addition to the Pew data.

Hurwitz, J., & Peffley, M. (2005). Explaining the great racial divide: Perceptions of fairness in the US criminal justice system. The journal of politics, 67(3), 762-783.

Peffley, M., & Hurwitz, J. (2010). Justice in America: The separate realities of Blacks and Whites. Cambridge University Press.

6. Finally, I would also emphasize that whereas the authors seemed surprised that they found "no consistent moderation by demographic group" (p. 16), they should not be. Prior research shows that procedural justice has similar effects across demographic groups (Wolfe et al., 2016). And, again, all the current authors are doing is testing the effect of a single dimension of procedural justice, what Mazerole et al. (2013, p. 35, 41) call demonstrating of "trustworthy motives."

This is another area where there are conflicting results from studies of procedural justice, and the authors' analysis adds to this discussion and in no way detracts from the argument of the work. I agree with their response.

Version 2:

Reviewer comments:

Reviewer #3

(Remarks to the Author)
The authors have address all my former concerns to my satisfaction. As such, my recommendation is to accept the paper in its current form for publication in Nature Communications. Congratulations!

Reviewer #4

(Remarks to the Author)
I appreciate the additional revisions the authors have made to the manuscript. The revised version better situates the work against past research and makes the contributions more legible. I have no further comments and look forward to seeing this work in print!

Reviewer #5

(Remarks to the Author)
In the Supplemental Material section Overview of Statistical Approach: BCF Analysis the authors describe the model for the Skin Conductance Levels only. This model appears to include person-level covariates only in the beta term (the "prognostic effect" term) and not in the tau term (the "treatment effect" term). This means that age, race, and gender cannot be inferred to moderate the treatment effects, only time. In this sense, this model is "simpler" than it could be, despite the authors remarking that this model was the "most complex" one in that in was for repeated measurements. Why were the x covariates not used in tau? Moreover, was this the case with the other models as well? I cannot discern because those models are not reproduced. Why not reproduce them all in detail and not just the SCL?

This is especially confusing in light of Table S8, which reports on what, if any, moderating effects there are on the other metrics. This table has outcomes on the columns and moderating variables (demographics, etc) on the rows. As far as I could see, there was no methodological explanation of how this table was produced. I was hoping that it was made using the BCF model, but that would only be possible if those demographic variables had been used, and I suspect they were not. As best I can tell, an ad hoc classical procedure was used to produce S8, though it was not (as far as I can find) described in any way, other than one mention of NHST? This should be fleshed out, as the claim of "no moderation" is an important one.

In my opinion, it would be nice to use the BCF model in the way it was intended, including x in the tau term and exploring the posterior of tau(x).

Version 3:

Reviewer comments:

Reviewer #5

(Remarks to the Author)
Thank you, that answers my questions and I believe will help other readers as well.

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Nature Communications **Reviewers' Comments:**

Reviewer #1 (Remarks to the Author):

I previously reviewed this paper. I appreciate the additional work on voice prosody and further description of the campus setting. I think the skin conductance findings show general arousal/intensity rather than anything more specific.

The authors have done an enormous amount of work on a fascinating topic. This is a strong paper that should be published.

For the purposes of this appeal, we wish to emphasize and appreciate the attention and work this reviewer has already undertaken throughout our revisions before considering it publication-ready.

Reviewer #2 (Remarks to the Author):

This study has serious shortcomings, which together mean that it adds little to the literature. Essentially, it uses a worse intervention …

Foreshadowing our replies below, the reviewer is calling a "worse" intervention one that is narrowly focused on one element (transparency of benevolent intentions) rather than a multi-pronged treatment that clusters many aspects of procedural justice theory and therefore is confounded with respect to specific elements. We believe this experimental precision allows us to make claims about a single concept, which is our goal. In our view, a precise experiment versus a confounded one is not better or worse, but simply serves different purposes and answers different research questions.

and worse samples that past experiments,

We believe that a sample should match its research question. We are not conducting an experiment to change policing outcomes (e.g., arrests). We are conducting an experiment to identify the impact of transparency statements in community policing interaction settings, which we designed after observing transparency statements in the field, following our qualitative analysis.

As we show below (in a table that summarizes every study we could find in the PJ literature) there has never been a study using a sample like ours to answer the questions we asked.

and just shows what we already know—namely, that exposure to procedural justice (here called "transparency statements")

Procedural justice involves many elements (i.e., dignity, trustworthiness, neutrality, and voice) and is not interchangeable with a transparency statement.

has a short-term, encounter-specific effect on attitudes toward the particular officer(s) in the interaction. To be crystal clear, the biggest limitations of the study are that it uses a weak measure of procedural justice (see below), calls it something different (see below), and then focuses only on specific (rather than general) policing attitudes. Let me explain further.

1. First, the authors have pulled out a single component (i.e., demonstration of trustworthy motives) from the broader procedural-justice package that has been tested repeatedly in much stronger field trials, after which they acted like it was something new, calling it "transparent communication of benevolent intent," without even seriously (or thoroughly) discussing the other trials.

First, our transparency statement intervention did not arise from the PJ framework. Rather, it emerged from a grounded theory approach that we detail in the qualitative Phase 1 component in the present manuscript. We observed hundreds of community policing interactions firsthand, and noticed how differently these interactions proceeded when the officer's benevolent intentions for the interaction were made clear immediately upon beginning the interaction. While the extant literature on PJ is consistent with some of the underlying psychology as to why our transparency statement intervention is effective (i.e. because it shares the belief that how officers treat community members in interactions matters), the intervention itself did not arise from the framework, so it is not accurate to characterize our intervention as emerging from existing literature. To further clarify these ideas in the manuscript, we have now revised the flow of the Introduction to foreground our qualitative observations before turning to a discussion of how our qualitative insights align with and diverge from existing PJ theory.

For example, on pg. 8 we now state:

> "We note that, though our treatment emerges from a bottom-up qualitative approach centered on nonenforcement interactions, our work shares a common belief with *procedural justice*—a prominent framework for improving the quality of community-police interactions in general: how community members are *treated* during interactions will matter for their trust in interactions with the police, and how legitimate they perceiver officers to be. Although these claims are well-supported in basic research, when procedural justice treatments have been implemented in real-world settings it has led to mixed results[4,5,6] and a high degree of unexplained heterogeneity ($I^2 = 95\%$; see also Table S2 in the SI)[1]. This high degree of unexplained heterogeneity suggests that research has not yet illuminated which contextual features predict whether PJ treatments will yield their intended effects (e.g., the type of interactions being studied). Thus, a key advantage of our bottom-up approach is helping us identify the potential impact of targeting the precise language officers used to initiate community policing interactions."

The literature is unequivocal that the linguistic demonstration of "trustworthy motives" is a central component of procedural justice (Mazerolle et al., 2013, p. 35, 41). Now, let me point to just a few of the trials that have tested procedural justice, only one of which is even cited (on p. 4). Mazerolle et al. (2013) did a field experiment (N = 2,746) randomizing the language officers used in actual traffic stops, a component of which was telling the civilians

who were stopped that they were not singled out and that the stops were important to save lives (i.e., a transparent statement of benevolent intent)…

We are aware that trustworthy motives are a component of procedural justice. Our contribution is different. We focus specifically on how to communicate trustworthy motives in community policing contexts, which the literature on PJ theory has not shown. A secondary benefit of our research is that we have clarified how to accomplish a key aspect of PJ theory's claims in community policing contexts. The studies referred to by the reviewer were not conducted in a community policing context, which was the focus of our research. Instead, they were in enforcement contexts, which are quite different and not the topic of our research.

We define transparency statements as the explicit communication of authentic, genuine, and personal intent to be benevolent (as opposed to harmful) in the focal interaction (not the community in general, or other groups). Importantly, this operationalization of transparency statements differs along several dimensions from prior operationalizations of trustworthy motives in PJ field experiments.

For example, the parts of the QCET mentioned above includes two statements:

- "In Queensland alone there were 354 deaths in 2009. One of the hardest parts of our job is to tell a person that their loved one has died or has been seriously injured in a traffic crash."
- "Can you please help us to reduce these accidents by continually driving carefully and responsibly?"

Neither of these statements communicate whether the officer's intent is benevolent and specific to that particular interaction (i.e., keeping the specific community member they are speaking to safe). These aspects clearly differentiate our work from prior operationalizations of trustworthy motives.

In the revised manuscript, we have further specified our transparency statement definition to further clarify this point. See pg. 6:

> "We call this a *transparency statement*. It involves a clear and truthful statement of the officer's benevolent intent for the interaction at the outset."

Similar field experiments were conducted by many others, such as Sahin et al. (2017), MacQueen and Bradford (2015), Terpstra et al., (2023), and Weisburd et al. (2022). And none of these experiments were conducted in the kind of artificial conditions used in the current study (e.g., telling participants they were in a study about natural interpersonal interactions, outfitting them with physiological sensors, and then having them have an odd conversation with an officer). That is, all prior field trials used actual police-civilian interactions with potential enforcement consequences (e.g., ticket, arrest).:

First, the conversations in Study 1 are part of the standard type of community policing conversation we observed in the field during our grounded theory phase. Furthermore, as we note in Table 1 in the main text, there were key advantages to consenting participants, namely that we could equip them with physiological sensors and audio record the conversations. This gave us

access to subtle, nonconscious outcome measures (i.e., physiological response, language) which would not be available in a natural field experiment.

Second, again due to our empirical context specifically being nonenforcement community policing interactions, we are not trying to speak to potential enforcement outcomes. However, it is worth noting that the supervisor of the on-duty officers who participated in this study directly communicated to these officers that they were to enforce the law if necessary, in the course of conducting the study. Therefore, though our focus was on community policing interactions—not enforcement interactions—the stakes of the interaction were the same as if these interactions were part of an officer's regular shift (see also the "Realistic Stakes" component of Table 1 in the main text).

Third, the QCET experiment did include a manipulation of officers' language, but these results were not nearly as definitive as suggested by the reviewer. A direct replication of the Mazerolle et al. (2013) QCET conducted by MacQueen & Bradford (2015) showed the exact opposite set of effects: "judgements of the procedural justice of the encounter and overall satisfaction *fell* in the experiment areas relative to the control areas" (pg. 432). Interestingly, the latter authors followed up in the department to do a post-mortem as to why the experiment backfired and found evidence of communication and implementation failures—specifically, that "the experimental units did not deliver the verbal messages with any consistency" (MacQueen & Bradford, 2017, pg. 333). While the authors conclude that these failures likely account for the discrepancy in results between their trial and that of Mazerolle et al. (2013), this is only a post hoc explanation. Our study provides significant advance over this pair of studies because it captures the exact language officers used in each interaction by audio recording each interaction, enabling us to conclude that officers did, in fact, utilize transparency statements at the beginning of the interaction when in the transparency statement condition, and that doing so had the pre-registered predicted effects on community member threat and trust.

In reflecting on the key differences between our study and those in the existing literature, we identified three critical elements in our study: an exogenously-administered treatment, an objective measure of officer behavior in community-police interactions, and firsthand measures of community member perceptions of the focal interaction.

(1) **An exogenously-administered treatment**: The most empirically rigorous way to determine the effect of an intervention is if it is administered as part of a research study (rather than up to officer discretion). This approach isolates the treatment and mitigates selection effects. Many PJ studies take a purely correlational approach, which means reverse pathways, confounding 'third' variables, and spurious relationships cannot be ruled out.

(2) **Objective measure of officer behavior**: To ensure that officers administered the treatment with fidelity, it is necessary to capture officers' actual language in each interaction. Without this data, if a treatment yields inconsistent, weak, or null effects, it is impossible to determine whether the treatment itself is not efficacious, or if it is something about how officers administered the treatment.

(3) **Firsthand measures of community member perceptions in the focal interaction**: Taken together with the prior two elements, the only way to

> systematically determine what officer behaviors can effectively reduce threat and increase trust is to obtain ground truth responses from the community members who participated in an interaction with an officer. Many PJ studies obtain only community members' *general* perceptions of the police, which cannot help identify specific levers that move the needle on community member perceptions and experiences.

We have conducted a literature review of $N = 110$ articles testing procedural justice theory, drawing from Bolger & Walters's 2019 meta-analysis, Nagin & Telep's 2020 systematic review, as well as additional literature review of articles that have been published since these two prior papers were released, which we now include in our SI (see Table S2). We coded each of these articles on the presence vs. absence of each of these 3 critical features of our studies.

Of the 110 articles, 95% (i.e., 104/110) omit at least one critical feature that our Study 1 includes. Of the 6 articles that do include all 3 features, 5 involve only video or survey vignette experimental manipulations, rather than manipulations in actual live interactions with community members as our Study 1 does.

The only study that investigates observer-verified officer behavior and firsthand community member perceptions obtains conflicting findings: observer-coded procedurally-just officer behavior improves as a result of PJ training. In contrast, there is no effect on community-member rated PJ of officers. This is in direct contrast to our findings where officer behavior (i.e., presence vs. absence of a transparency statement) directly predicts community-member rated threat and trust.

The results of this review further clarify the contribution of our research both theoretically and methodologically: the high internal validity of utilizing a well-controlled naturalistic study in the field, as we do in Study 1, yields consistent evidence across all metrics collected (i.e., community members' self-report, physiological, and linguistic responses).

Relaxing the assumption of achieving all 3 critical features to any paper that employs an experimental treatment yields 39/110 papers. Of those 39, 21 include no direct, objective measures of real officer behavior (i.e., 2 rely on the presence vs. absence of a body-worn camera as a proxy for procedurally-just behavior, 7 rely solely on community member survey ratings of officer behavior, and 10 rely on only hypothetical survey or video vignettes—1 of which utilizes actors rather than real police officers); 5 include experimental treatments that were broader than purely PJ, and 7 do not include firsthand measures of community member perceptions.

Taken together, no research to date has tested the impact of an experimentally precise manipulation of one linguistic element (i.e., transparency of benevolent intentions) rather than a multi-pronged treatment that clusters many elements of PJ theory. Moreover, no studies have tested a manipulation with this precision at the outset of interactions. Our results suggest that a singular treatment at the outset of an interaction is sufficient to meaningfully reduce threat and increase trust. Notably, this treatment is pragmatically much simpler for officers to remember and employ in their interactions, compared to the lengthy scripts employed in previous trials (which introduces another potential source of implementation heterogeneity).

2. Second, beyond the field trials discussed above, there is a huge body of survey experiments that basically do the same thing (but with better interventions and better outcomes) than the current study does in experiments 2 through 7 (e.g., Johnson et al., 2017; Lowrey et al., 2016; Maguire et al., 2016; Roche, 2019). That is, the last six experiments in the current study look very similar to those that appear in mid-level special journals, and really add nothing new over the findings in those mid-tier articles.

Again, the survey experiments mentioned by R2 above do not show transparency statement effects. They come directly from the PJ approach and aim to test the effect of PJ more generally. While some do begin to pit different elements of the PJ framework against one another to determine the relative magnitude of effects, none of them directly test an intervention that maps onto our transparency statement. Moreover, our studies come from a different tradition: they began with a grounded theory approach, leading to the transparency statement hypothesis that we test across 7 studies. Study 1 offers a fundamental test of the key treatment in the field, and then Studies 2-7 precisely test the features of the transparency statement that must be present to effectively reduce community member threat in encounters with police. Our research does not solely rely on vignette studies—it includes both direct firsthand qualitative observations of real community police interactions in the field as well as a naturalistic study in the field. Our naturalistic study in the field includes novel outcome measures of participant behavior that have never been tested before in the literature on community-police interactions: both their physiological response via skin conductance level, as well as their language in the interaction. In this way, the study package as a whole offers significantly more specificity than other experiments that rely exclusively on vignettes.

Furthermore, given our focus on community policing contexts, it is not clear what "better interventions and better outcomes" would mean. The community policing context is simply different from other interaction contexts where PJ is typically studied. Another key benefit of the grounded theory approach we employed is to ensure that our intervention and outcome measures are tailored to test the efficacy of our intervention in the community policing context.

3. Third, and this is very important, the key finding from past experimental work—both from the field trials and survey experiments—is that procedural justice interventions, like transparency statements demonstrating trustworthy motives (or benevolent intent), mainly affect encounter-specific attitudes, but have little effect on general policing attitudes (e.g., Johnson et al., 2017; Lowrey et al., 2016; Maguire et al., 2016; Sahin et al., 2017). Nagin and Telep (2016, p. 10) emphasize this point: "There was a significant impact only on these encounter-specific outcomes and not on more general views about the police." This is a huge problem for the current study where the entire focus is on encounter-specific attitudes and little attention is given to general policing attitudes. That is, in the current study the key outcomes are perceptions of the specific officer in the interaction (e.g., p. 10: "The officer just wanted to make friendly conversation"). The outcomes are NOT general policing attitudes. This is true even in the later experiments where some questions ask about the specific officer or "another officer from this organization" (e.g., how likely to "wave if you see this officer", p. 30). So, what the study basically shows is that if an officer explicitly says they just want to make friendly conservation (e.g., "I'm just out and about walking around, talking to people in the community"), then civilians are more likely to

believe that the specific officer is just trying to have a friendly conversation. Although this is interesting, it is not groundbreaking.

We are not questioning the overall theory of change of community policing in our research. Community policing's theory of change is that many interactions over time will cumulatively rebuild trust. However, a necessary precondition before research can reach the point of studying the impact of many interactions over time is to provide strong evidence supporting the efficacy of communication strategies that make a single interaction go well.

Therefore, the purpose of our study does not rest upon testing the link between encounter-specific and global attitudes. Nevertheless, we deeply engaged with the PJ literature to understand this potential critique. Our review of the PJ literature led us to conclude that the evidence for the link (or lack thereof) between encounter-specific and global attitudes is mixed:

- Some experimental studies show inconsistent effects even on encounter-specific attitudes (e.g., Langley, 2014; MacQueen & Bradford, 2015).
- In contrast, some experimental studies *do* find consistent effects on encounter-specific and global attitudes. For example, Maguire et al. (2017) find significant encounter-specific *and* global effects of a procedural justice video manipulation (though global effects are smaller in magnitude). Similarly, Mazerolle et al. (2012; 2013) in the QCET find significant effects on participants' self-reported belief that their view of police *in general* has changed as a function of the experimental treatment. Peyton et al. (2019) also show improvements in general attitudes 3 weeks later. Finally, Trinkner (2019) shows general effects on PJ of a single forum session for parolees on general attitudes.
- Many studies cannot even speak to a possible distinction because they *only* collect encounter-specific measures (i.e., do not even collect global attitudes, e.g., Roche 2019).
- Research not specifically on police contact and PJ but more generally on intergroup contact (i.e., contact with a disliked/distrusted outgroup) further shows a linkage between a specific encounter and people's future contact with members of that outgroup (e.g., Meleady & Forder, 2019).

In summary, these facts in the literature contradict the reviewer's suggestion that our study is far inferior to a PJ literature that regularly establishes effects on general attitudes. Instead, our study's focus on specific attitudes matches the majority of the PJ literature, and therefore should not be viewed as any less of a contribution.

4. I would also emphasize that on page 4, the current authors are misrepresenting the literature when they claim that the results of past work are mixed. They cite Bolger and Waters (2019), but that meta-analysis concluded that there was strong support for the theory of procedural justice: "The analysis produced results in support of all three causal chains of Tyler's theory … Citizen perceptions of police procedural justice would appear to have both a direct impact and indirect effect on citizen cooperation, the latter by way of police legitimacy beliefs" (p. 98)…

We do not disagree with the main effects in the meta-analysis. We are simply pointing to the high degree of unexplained heterogeneity, which illuminates that research has not yet identified which

contextual features predict whether PJ treatments will yield their intended effects (e.g., the type of interactions being studied). Meta-analysis experts routinely and consistently suggest that main effects are not meaningful in the context of heterogeneity; on this point the statistical experts are clear (see, e.g., [7,8]). When effects are so heterogeneous, it suggests the construct is not well-understood or predictable and therefore unlikely to consistently improve outcomes in the real world. Thus, our study's focus on specific language could help address some of the unexplained heterogeneity.

That meta-analysis also was not focused on experiments or field trials, but mostly included cross-sectional surveys looking at associations between perceptions and beliefs; so it is not really relevant here, where the specific focus is on experimental effects of interventions, both in the current study and in the relevant past work (e.g., Johnson et al., 2017; Lowrey et al., 2016; MacQueen & Bradford, 2015; Maguire et al., 2016; Mazrolle et al., 2013; Sahin et al., 2017; Weisburd et al., 2022). What a review of all the experiments shows is that they only meaningfully affect encounter-specific perceptions, just like the current authors find.

In our revised introduction, we now describe the literature on PJ as having many studies showing main effects, but also a great deal of unexplained heterogeneity, including failed direct replications of some of the literature's most prominent studies (e.g.,[9]; see pg. 8 in the main text).

As noted, we also conducted a literature review of $N = 110$ articles testing procedural justice theory, drawing from the Bolger & Walters meta-analysis, the Nagin & Telep systematic review, as well as additional literature review of articles that have been published since these two prior papers were released. In our $N = 110$ article review, we specifically delineated whether each study included an experimental treatment, and whether the study yielded consistent effects. The results of this review reveal that 44% of the articles that included an experimental treatment (i.e., 17/39) yielded inconsistent effects or effects that run counter to the predictions of PJ theory (1 of which has been retracted since being published). We believe that the high percentage of experimental studies showing inconsistent or counter effects clearly justifies labeling the results of this literature as mixed.

5. Shockingly, the authors say on pages 15-16 that "people of color and those from lower-SES contexts might have more negative attitudes toward law enforcement" and then they cite a Gallup poll. This is not a "might have" issue. There is a tremendous amount of very strong research that shows quite clearly that such Americans do have more negative policing attitudes and emotions (e.g., Pickett et al., 2022).

In Study 2, we wanted to present the logic behind our theorizing to help explain our competing predictions about potential moderating effects. While research suggests that people of color and those from lower-SES contexts do, on average, have more negative attitudes towards law enforcement, we did not want to definitively presume this would be the case in our data a priori. Nevertheless, we agree that we can state more definitively what previous research has found, while still couching our predictions utilizing more cautious language.

<u>As such, we have now revised the writing and citations around these statements in the revised manuscript, specifically including the Pickett et al. citation (see pgs. 15-16):</u>

> "On the one hand, people of color and those from lower-SES contexts tend to have more negative attitudes towards law enforcement at baseline."

We also independently reviewed the Pickett et al., 2022 research cited by the reviewer, and drew different conclusions. For example, in Figure 1, reproduced below, a majority (i.e., 55%) of Black Americans do indicate being afraid or very afraid of being killed by police (denoted in the two righthand most red bars). While this is a higher proportion than White Americans (~15%, denoted in two righthand most blue bars), from a scientific writing perspective, it is inaccurate to make claims about Black Americans' universal attitudes towards police officers with absolute certainty, because 45% of Black Americans did not feel this way.

[figure redacted]

6. Finally, I would also emphasize that whereas the authors seemed surprised that they found "no consistent moderation by demographic group" (p. 16), they should not be. Prior research shows that procedural justice has similar effects across demographic groups (Wolfe et al., 2016). And, again, all the current authors are doing is testing the effect of a single dimension of procedural justice, what Mazerole et al. (2013, p. 35, 41) call demonstrating of "trustworthy motives."

We added this study and these moderation results specifically to respond to a direct request of a Reviewer in a prior round of review. Specifically, this reviewer requested we add a study that was adequately powered to explore potentially heterogeneous effects by participant demographic group membership.

Our prior response to this prior Reviewer explained: A key motivator of this work was the record low trust levels in law enforcement, and our own field observations of how just how well interactions could go with historically over-policed populations, when enacted transparently. As such, we have now taken additional steps to test the efficacy of our transparency statement intervention in a large (N = 969), in a sample of U.S. adults that was constructed to resemble the U.S. population's marginal distributions of age, gender, and race/ethnicity, so that we would be adequately powered to detect any evidence of moderation by racial group membership.

Specifically, we partnered with active law enforcement officials to create video stimuli that we could record and use in an online experimental test of the transparency statement intervention in a large ($N = 969$) sample that was constructed to resemble the U.S. population's marginal distributions of age, gender, and race/ethnicity. Just as we did in the field experiment, we trained the partner officer on what a transparency statement would consist of, and co-created with him a statement that would feel genuine and authentic to him, were he to engage in a nonenforcement interaction on the job. Proceeding from this, we video recorded him introducing himself twice: once, as he normally would (control condition), and a second time following the tenets of the transparency statement intervention ("I'm just trying to get to know my beat, and my community"; see new Experiment 2). The results now appear in the manuscript.

Therefore, we do believe it is appropriate to present the lack of consistent by demographic group moderation results. We believe it is appropriate to label it as surprising given extant research for at least three reasons. First, Wolfe et al. do not show purely invariant results—in fact, they document that the effect of procedural justice on trust is *stronger* for those who were the victim of a crime in the past 6 months. This is critical because people of color and lower-SES individuals are more likely to experience victimization than their racial majority and higher-SES counterparts[10]. This finding itself casts doubt on whether there is clear evidence in the PJ literature that effects on trust are similar across demographic groups.

Second, as we outline above in our response to the prior Point #1, we are not simply testing the effect of a single dimension of procedural justice, so it is not clear whether previous studies explicitly seeking to test the broad theory of procedural justice are applicable or reduplicative with our findings.

Third, even if our findings were intended to test PJ theory specifically, it is not clear that doing so would not be novel. Wolfe et al. (2016) investigate general procedural justice and moderation by demographic group, so it does not immediately follow that these results would extend or replicate to our studies, where we were assessing interaction-specific threat and trust (since as noted in the earlier Point #3, there tends to be divergence between encounter-specific and global attitudes results).

Moreover, more recent studies than Wolfe et al. (2016)—including the meta-analysis we previously cited[1] and a brand-new meta-analysis specifically on social identity groups and PJ[11]—have called into question the invariance thesis and found significant evidence of variation when utilizing within-person longitudinal methodologies[12], motivational measures of postures towards police (e.g., compliant vs. defiant)[13], citizens' felt stigmatization[14], and citizens' propensity to trust the police[15]. Indeed, a recent study that directly sought to replicate Wolfe et al.'s (2016) findings only found partial support: they found that procedural effectiveness perceptions mattered more for those with ethnic minority (vs. majority) identities[16].

REVIEWER 2 REFERENCES

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Reviewer #3 (Remarks to the Authors):

This is my second review of the current manuscript (I previously reviewed an earlier version in another journal). Overall, the paper has been significantly improved, mainly because of certain edits and added empirical evidence. In this sense, I thank the authors for being responsive in the way they addressed most of my formerly raised concerns. That said, I think there are several issues remaining that warrant further attention, and I am still not convinced that the current piece meets the extremely high bar for publication in a journal like *Nature Communications*. Unlike my former review, however, I am willing to opt for a major revision recommendation this time, although I perceive such a potential revision as risky. Below are my remaining main concerns and recommendations to the authors.

1. **While some people believe that "words are violence," I personally think there is an important distinction to be made between physical actions and language, which partially motivated my former comment about the lack of clear behavioral evidence in the field study. That does not mean, as the authors seem to imply in a somewhat passive-aggressive tone in their revision notes, that I do not see the harm caused when someone psychologically abuses another individual. Still, the authors have conducted a study on a college campus, with university students likely constituting a large share of their sample, and with all participants being aware of that they took part in a study. Moreover, the primary outcomes are either self-reported, physiological, or language-based variables. I maintain that this limits the behavioral implications that can be drawn from the study in question and its assumed generalizability to the general population.**

We have now changed the implications and limitations throughout the entirety of the General Discussion to ensure we do not overclaim. We do not assume generalizability from Study 1 but instead test for generalizability using the national samples in Studies 2-7, while acknowledging limitations from using an online study methodology to obtain national samples.

<u>To provide just one representative example, on pg. 19 in the General Discussion, we now state:</u>

> "Our work was conducted utilizing a variety of distinct U.S. samples (observations in large urban areas, naturalistic field study in large, urban area, and national online samples). However, future work should continue to extend and replicate our findings in different communities to expand the generalizability of the transparency statement intervention."

> **I was also not convinced by the way the authors addressed this concern, and a related one posed by Reviewer 1 (comment nr. 2), regarding certain labels (i.e., "community members") and issues related to generalizability and representativeness. With respect to the label of "community members," would the authors feel equally confident in referring to university students or online panel members as consumers, patients, or any other generic group simply because virtually all students and online panel members are, or have once been, consumers, patients, or something else? My point is not that the field study is *in*valid; rather, my point is that the authors appear to continue "overselling" the field study, including**

> **its sample and focal outcomes. What they have done is impressive, labor-intense, and extremely difficult, but this work does not need to be extrapolated beyond its core content (which might well be impressive enough on most points). I understand the authors' opinion that language can be thought of as a behavioral outcome. But if I were to use the authors' own style of posing rhetoric questions (which they seem to do in the revision notes when they ask whether verbal abuse is farther away from a behavioral outcome than physical abuse), then wouldn't it also be possible to label the hand and finger movements needed to reply to survey items in a questionnaire as a form of behavior, if an equally extreme position would be taken in the opposite direction? Few scholars would agree that hand and finger movements constitute compelling behavioral evidence. I agree that language and interpersonal interactions are closer to behavioral outcomes (Baumeister et al., 2022), but I still think "actions speak louder than words." To me, language is often goal-oriented and hence used either to attain rewards or avoid punishments (except for small talk). In other words, language may well guide certain actions, but I view these actions as real, observable behavior more so than I view the words uttered to approach certain goals or avoid threats.**

We agree that it is advisable to ensure that the conclusions of our work are interpreted appropriately in light of the evidence we have obtained. As such, throughout the manuscript, we have toned down our language regarding behavioral outcomes and generalizability of our findings through rewriting. We believe that these revisions to the writing present our findings to readers with the appropriate level of caution.

To present just two examples, on pg. 12 we now state:

> "This difference is *potentially* meaningful because it reveals that participants were using the type of language that *tends to* form the foundation of positive rapport that makes people feel unthreatened and open to trust."

Similarly, in the General Discussion on pg. 20 we now state:

> "Our study showed that absent a transparency statement at the start of the conversation, the officers' questions contributed to a sense of threat. *Potentially supporting this threat*, in these interactions community members also displayed an avoidant conversational style (i.e., less authentic, more terse) and physiological profile (lower initial arousal). When officers made transparency statements in the first two sentences of the conversation, however, it changed the tenor of the ensuing interaction. For example, *an exploratory analysis* revealed that twice as many community members reported feeling *inspired* by the end of the interaction when officers made a transparency statement (28%) than when they did not (14%; see SI and Figure 1C). *There was evidence consistent with* this shift in transcripts, self-reports, and stress physiology."
> **emphases added to indicate where language was softened*

Nevertheless, while we do feel it is a worthwhile scientific endeavor to understand what constitutes behavior in general in social science research, this is beyond the scope of our research. Our research began squarely from observations of interpersonal interactions in the community policing context. Indeed, the key objective that police agencies and community

members care about in this context is engaging in more positive, trusting interactions. We index "positive, trusting interactions," via language, physiology, and community members' post-interaction ratings. These indicators have been validated in the literature on interpersonal interactions (e.g., in interracial interactions[17]; in conflict interactions[18,19]; in romantic relationships[20]). Across these three indicators, we obtain consistent results pointing to the idea that these interactions were more positive and trusting when officers began with a transparency statement.

Finally, we did not mean any disrespect with the analogy about verbal and physical abuse, but rather we simply meant to make our point more clearly with a concrete, high-stakes example.

2. **In their revision notes (p. 6), the authors present a table featuring the local police department 2021 community survey results. They then use the percentages in the table to argue for their position that no meaningful differences exist in the prevalence of certain crime types on-campus versus off- campus. I wonder what the "Other" category represents in this table. If, for example, that category would include murders and/or homicide, then even a 5 percent difference between on-campus and off-campus sites would have implications for life and death, which seems to go against the authors' claims of no meaningful differences between on-campus and off-campus settings. Relatedly, I do not agree with the authors that there are no available studies demonstrating the importance of location on people's responses in a way that might have influenced the field results. For instance, it has been known and shown for half a century that background authority matters, with about two-thirds of participants in Milgram's (in) famous obedience research complying with giving others "lethal" electric shocks when the data were collected at (the prestigious and authority-signaling) Yale University, whereas compliance rates went well below 50 percent when the same study setup was replicated at a (rundown and substantially less prestigious) commercial building located in the downtown shopping area (Milgram, 1974). Thus, I agree with Reviewer 1 that the setting of the field study is problematic if the authors want to draw generalizable conclusions that go beyond the specific sample included in this study, particularly if such conclusions extend to behavioral implications.**

Our purpose in referencing the community survey results was not to say that on-campus and off-campus contexts are identical, but rather to say that there is likely less of a distinction in public safety concerns in this particular college campus setting than might be the case in a college campus that is located far away from an urban center.

However, it is certainly true that the context of any field study would not readily be generalizable to all communities everywhere. We have addressed some questions of generalizability by including evidence from both our initial qualitative field observations and our subsequent studies—particularly Study 2 which was constructed to resemble the U.S. national population in terms of age, gender, and race/ethnicity—providing initial evidence that our findings would generalize beyond the specific community where Study 1 took place. Nevertheless, we hope that the initial findings we present here will spur on many future research investigations that seek to

replicate and extend the current findings in other communities to provide the strongest evidence of broad generalizability.

Therefore, we have added this as a limitation and direction for future research to the General Discussion on pg. 19:

> "Our work was conducted utilizing a variety of distinct U.S. samples (observations in large urban areas, naturalistic field study in large, urban area, and national online samples). However, future work should continue to extend and replicate our findings in different communities to expand the generalizability of the transparency statement intervention."

3. **Sticking to the field study, the authors state in their manuscript that it "was a precursor to a future program of field research [...] that could test the long-term effects of transparency on public safety" (p. 7). If such long-term oriented work now exists as available data, which could counter many of the raised concerns in this (and the former) round of reviews, I strongly recommend the authors to include the described long-term effects in the current package of studies rather than as a separate submission elsewhere. Doing so would materially strengthen the confidence in the authors' findings, not only from a short-term horizon, and would further underscore the societal relevance of their research.**

Our key contribution in this research was to first discover the concept of transparent communication of benevolent intent in our qualitative observations in the field. From these field observations, we then derived a hypothesis about how to test the transparency statement in a well-controlled study that simulated real community policing interactions as closely as possible. This study is needed first because it is necessary to identify that the linguistic strategy indeed makes a single interaction go well before it is possible to scale up to the long-term effects that many interactions would have over time (e.g., broader indicators of public safety).

Furthermore, we have included all the data and studies that we have conducted in this line of research on transparency statements to date in the current study package. We are not withholding any additional field data in hopes of publishing a separate submission elsewhere. In contrast, it is our firm belief that we needed to put this initial paper through the review process first before we could in good faith bring a police agency on board to institute transparency statement training to all of their sworn officers. We want to be confident that our findings have been peer-reviewed before we bring our transparency statement training to an entire agency.

4. **In their revision notes, the authors also state that they are not aware of any studies showing a discrepancy between the findings obtained from lab experiments (where participants know they take part in a study) and field experiments (where participants typically do not know they take part in a study). While there is a positive correlation between findings from lab and field settings at the aggregate level, it is also true that this correlation is substantially weaker in some disciplines, including disciplines that the authors of the current manuscript rely on. For**

example, Mitchell (2012) reported that 21 of 80 (26.3%) laboratory effects from social psychology switched sign between research settings (lab vs. field).

We are researchers who have been trained in the field of management and organizational behavior—which is labeled as "I-O psychology" in Mitchell's (2012) review. Importantly, Mitchell finds that the highest field-lab correlation comes from I-O psychology: $r = 0.89$. Therefore, while we agree that there are certainly contexts and sub-disciplines where laboratory effects differ from field effects—particularly in social psychology—our approach, which maps onto the approach more commonly taken in management studies, does tend to yield consistent effects across lab and field. In further support of this, our findings are consistent in their direction (and general magnitude) in a field-based setting and more acontextual settings of online experiments. One critical reason we believe our findings do not suffer from a field-lab decoupling is due to the grounded theory approach we took before undertaking any experiments. Observing many real community-police interactions in the field before conducting our experimental phase enabled us to maximize realism in both our naturalistic study in the field and our online experiments (e.g., using a real, uniformed police officer and his real introductions for video recording stimuli in Study 2).

5. **Regarding my formerly raised point about timing (in the last round of reviews), the authors seem to refute this issue as a meaningful confound by splitting the control condition into three smaller groups (n = 10, n = 11, and n = 109, respectively) depending on whether officers made a spontaneous transparency statement, an elicited transparency statement, or no transparency statement at all. Specifically, they present three p-values of p = .05, p = .10, and p = .40 as evidence against the idea that the timing and nature of the transparency statement could have influenced the results. With or without a Bonferroni correction, I do not think these results and the corresponding p-values provide compelling evidence for the absence of effects, considering the extremely small cell sizes in two of the three groups coupled with the large absolute mean differences between the groups. If the authors decide to retain this position, then at a minimum they need to supplement the (statistically non-significant) p-values with effect sizes.**

We apologize for not including effect sizes alongside the *p*-values we presented in the last round of review.

<u>We have now added effect sizes to the analysis of transparency statements in the control condition, see below and pg. 33 in the SI:</u>

> "Officers used their discretion in the later parts of the conversation, 7.7% of conversations in the control condition ($n = 10$), were coded for the presence of a spontaneous transparency statement, and another 8.5% of conversations were coded for the presence of an *elicited* transparency statement ($n = 11$), whereby the civilian directly asked the officer a question that led them to issue a transparency statement. However, exploratory analyses revealed that the presence of a spontaneous or elicited transparency statement later in the conversation did not have an effect on our key pre-registered

> outcome measures of perceived transparency ($p = .10$, $\eta^2 = .04$), threat ($p = .40$, $\eta^2 = .01$), and trust ($p = .05$, $\eta^2 = .04$)."

Though these effect sizes may appear somewhat large, they need to be interpreted with caution. This is because the observational nature of this analysis cannot account for endogenous correlations. For example, in the transcripts of interactions where a transparency statement was elicited, the community member actively questioned the officer as to their intentions, perhaps because they felt threatened. Similarly, in the interactions where an officer spontaneously issued a transparency statement later in the conversation, it is also possible that the officer did so because they sensed, for example, that the community member felt threatened.

Second, the reason we did this coding and analysis in the last round of review was to provide initial evidence as to why the transparency statement needs to occur at the outset of the interactions (rather than later on). To further substantiate this claim, we now reference research that has been published since our research was last under review, which provides a strong rationale as to why transparency statements are needed immediately at the outset of interactions[21].

Olivett & March (2023) conduct cognitive psychology experiments where participants were briefly exposed to visual stimuli of uniformed police officers (vs. non-officers). They obtained participants' in-the-moment physiological responses to mere exposure to an officer (vs. non-officer) stimulus. Across three distinct instantaneous physiological response measures (i.e., avoidance response latency, freeze response, and startle eyeblink amplitude), they find consistent evidence that civilians immediately evince threat and freeze responses in reaction to police (vs non-police) stimuli. These universal and immediate threat responses speak to the utility of intervening as we do at the outset of interactions between police and community members.

Moreover, in the General Discussion and implications section of their work, the authors suggest that intervening at the outset of community-police interactions is likely to be effective: "Police-threat associations may evoke defensive behaviors that are difficult to control or inhibit during an initial split-second expression. The effects observed in the current work were on a magnitude of milliseconds, millimeters, and millivolts. *Yet in situations where people are making split-second decisions—such as in police-civilian encounters—small effects can have large ramifications*.… Focusing on contextual features known to provoke defensive responses might be *one avenue for mitigating this harmful process before it begins*." (*italics* = emphasis added)

We now discuss the findings of this paper in the Introduction and explicitly reference it when discussing the effects of timing in the General Discussion on pg. 19:

> "Second, our work began with a qualitative observation phase which led us to conclude that transparency statements worked best in the field when they came first. These observations inspired our hypotheses and the design of our field experiment, leading us to intervene on first contact. Our observations are further supported by related basic psychophysiological research that documents an immediate civilian threat response upon mere exposure to police (vs. non-police) stimuli. Furthermore, supplemental analyses of our Study 1 (see SI) indicate that for the few conversations in the control condition where

an officer made a transparency statement later in the interaction, there were no observable benefits relative to those conversations where no transparency statement was made, providing initial exploratory evidence that transparency statements are most beneficial when they come first. Due to our field observations and recent basic psychophysiological research, our theorizing rested on the effect of transparency statements at the outset of the conversation (vs. not). However, future work should systematically examine whether transparency statements issued later on in conversations confer any benefits."

6. **When it comes to my former question about interrater reliability for the qualitative work (which Reviewer 1 also seemed to comment on), the authors appear to think that I confused the grounded theory approach with a content analysis. I did not. Moreover, the authors seem to imply that my question does not "rhyme" well with established methods for grounded theory, seemingly inferring either that the reliability of the coders is irrelevant to their work or that research based on grounded theory does not include interrater agreement scores. I firmly believe that reliability is important in any scientific investigation (although not always the most important aspect of academic work), and I also note that it is not uncommon to report interrater reliability coefficients in studies using the grounded theory approach (e.g., Dawes & Larson, 2011; see also Armstrong et al., 1997; Malodia et al., 2021).**

While it is not necessarily incorrect to provide interrater reliability statistics, up-to-date methodological guides to grounded theory approaches do not typically recommend reporting them[22]. This is because the fundamental tenet of a grounded theory approach is iteratively and continuously coding the data while continuing to collect additional data until reaching theoretical saturation[23]. Therefore, traditional hypothetico-deductive statistics do not accurately capture the coding process undertaken to analyze data collected using a grounded theory approach, and such statistics are not utilized to verify that an author team has appropriately used grounded theory.

Instead, up-to-date practices recommend that authors explain the process they undertook while collecting and analyzing the data, as we do in the methods section on pgs. 22-23 (key steps adhering to a grounded theory approach are italicized):

"Field notes and interview transcripts were analyzed by the two lead researchers by *discussing the day's data collection immediately or soon after it finished, and writing informal, conceptual, theoretical memos to summarize the emerging themes.* Transcripts and notes that were flagged as indicative of key themes were then analyzed line-by-line to understand recurring patterns and to check assumptions against the data. *Once the key themes were identified and refined (presented in Box 1), then exemplary quotations were selected to illustrate those themes to readers*. These quotations were then vetted with a third researcher, to reach agreement on the fit between the data and our interpretations of the data. *After reducing the qualitative data to the key themes and focal examples (Box 1),* we used these insights to design the randomized experiments and test the hypotheses directly."

7. **In my former review, I also explicitly asked the authors to clarify whether there are any arguments for the idea that aggressive policing would *not* generate higher threat perceptions than the transparency and control conditions. The authors seem to have left this comment largely unaddressed by simply disagreeing (without providing any clear justification for the disagreement). I expect that they address this point in case the Editor opts for the revision decision.**

In the last round of review, we sought to address concerns particularly about the evidentiary value of the experimental design in Experiment 4. In the revised manuscript, we have sought to further clarify our rationale for Experiment 4 and reframe the results around this clarified rationale. Specifically, this study explicitly manipulates the *benevolence* component of our transparency statement definition. By creating two conditions that both include an explicit statement of intent, and varying only the benevolence of that intent, we are able to empirically demonstrate that only when transparency statements communicate benevolent intent do they reduce threat, relative to a control condition.

<u>See our revised description of Study 4 below (also see pg. 17):</u>

> "We next asked: Do officers' intentions need to explicitly convey *benevolence* to have a beneficial effect? In Experiment 4 (N = 382), we used a three-condition design to compare the transparency statement to a new introduction that was transparent about the officer's intentions, but the intentions conveyed a potentially harmful intent (i.e., the officer is "trying to find a suspect in the area"), and the control condition. We suspected that the new introductory statement—though transparent—would not be as effective because it does not disabuse the participant of the significant concern that they will be accused of a crime. We found that the introduction that conveyed potentially harmful intent was highly different from the replicated reduction in threat from the transparency statement, Difference in ATEs = -0.46 *SD* [-0.61, - 0.33], pr(ATEambiguous> ATEtransparency) > .99. Furthermore, the introduction that conveyed potentially harmful intent also substantially increased threat compared to a control condition, ATE = 0.39 *SD* [0.25, 0.53], pr(ATE>0) > .99, consistent with qualitative accounts of the investigatory stop in the proactive policing model. Thus, only transparency statements that explicitly convey benevolent intent reduce threat."

Reviewer 3 References

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REVIEWER COMMENTS

Reviewer #3 (Remarks to the Author):

This is my third review of the manuscript, entitled "A Transparency Statement Improves Community-Police Interactions." I was asked to assess whether my formerly raised concerns had been adequately addressed and whether the issues mentioned by Reviewer 2 from the previous round of reviews had also been satisfactorily handled. In my view, the authors have done a tremendous job, and I am favorable toward publication of the current piece if my remaining issues, summarized below, are carefully addressed with corresponding edits in the manuscript and its supplementary materials.

We are grateful for these comments, we appreciate the careful review of our revised manuscript. These additional comments have helped us further strengthen the manuscript.

1. Regarding the claim, "exploratory analyses revealed that the presence of a spontaneous or elicited transparency statement later in the conversation did not have an effect on our key pre-registered outcome measures of perceived transparency ($p = .10$, $\eta 2 = .04$), threat ($p = .40$, $\eta 2 = .01$), and trust ($p = .05$, $\eta 2 = .04$)" please support it with equivalence testing using Bayesian statistics and/or, if applicable, the frequentist equivalence testing approach using the Two-One-Sided-Test (TOST) framework. Also, two of these three effects resemble the typical effect sizes in psychological science (e.g., Funder & Ozer, 2019; Gignac & Szodorai, 2016; see also Götz et al., 2022), which does not seem to provide compelling evidence for the absence of effects. At least this fact deserves to be acknowledged somewhere. Let the reader decide whether these null findings are interpreted as sufficiently strong evidence for the absence of a given effect.

We are grateful for the suggestion to conduct these additional analyses. We acknowledge that these results are exploratory and should not be interpreted. As recommended, we have conducted Bayesian equivalence analyses and added the writeup of these results in the SI.

Therefore, when we report these results in the revised SI, we now add sentences cautioning readers that more work should be done (see pg. S35):

> **"Exploratory analyses revealed that the presence of a spontaneous or elicited transparency statement later in the conversation yielded nonsignificant effects on our key pre-registered outcome measures of perceived transparency ($p = .10$, $\eta 2 = .04$), threat ($p = .40$, $\eta 2 = .01$), and trust ($p = .05$, $\eta 2 = .04$). Bayes Factors (B) using the BayesFactor package in R were conducted to corroborate NHST findings. Following best practices, $B < 1/3$ indicates moderate evidence for the null model over the alternative model. A B between 1/3 and 3 is inconclusive, and a $B > 3$ indicates moderate evidence for the alternative model over the null model. There was evidence for the null model for threat (threat $B = 0.302$). Results were inconclusive for transparency and trust (transparency $B = 0.667$; trust $B = 1.599$). Though failing to reach conventional levels of significance, we note that these analyses were exploratory in nature and relied on relatively small sample sizes. Therefore, we caution against interpreting these results, and future research should**

seek to systematically explore the effects of spontaneous and elicited transparency statements that occur later in community-police conversations (vs. at the start).

2. In my former review, I noted that Mitchell (2012) found 21 of 80 (26.3%) laboratory effects from social psychology to switch sign between research settings (lab vs. field). The authors seem to somewhat explain away this fact despite that they draw on findings that are at least partially based on social psychological literature by simply claiming that they are scholars in management and organizational behavior, with this field conveniently suffering less from results discrepancies between lab and field studies according to Mitchell (2012). While I buy the authors' arguments, I think they can provide some more nuance by noting that portions of the literature they draw on are contextually sensitive (Elbæk et al., 2023; Van Bavel et al., 2016), although they, themselves, find consistent evidence for their theorizing.

Thank you for this suggestion. We agree this is an important point worth noting. We have outlined this as a point for future research and incorporated the relevant citations you provided in the revised General Discussion section on pg. 20:

"Third, our mixed-methods approach yielded consistent findings across both relatively naturalistic settings in the field and controlled, online settings, consistent with meta-analytic reviews of organizational behavior research. However, of note, not all social science disciplines have such high correlations between field and laboratory settings (e.g., social psychology, r = 0.53). Indeed, many social-psychological effects are contextually sensitive. Therefore, future research should seek to understand more deeply the parameters that lead to consistent vs. inconsistent effects in the field and the laboratory."

3. Regarding Table S2, I think the authors can do more to show how their own research is unique and differs from most other investigations. I suggest that they add their own paper either in the top or bottom row of the table, while also including more columns such that it becomes conspicuously clear that their own work does much more than what is typically the case in most prior related research. Some suggested columns that can be added are main theory, preregistration, open data, mixed-methods investigation, and physiological measures. Obviously, such a table means that the authors would check all boxes, while most (if not all) previous articles would only check a subset of these boxes (for a similar approach, see Andrews et al., 2014).

We very much appreciate this suggestion to further highlight how our own research differs from our review of previous research in this domain. We have added our paper to the top row of Table S2 (see pgs. S5-27). We also appreciated the suggestions for how to utilize the table to further highlight the key novel and differentiating aspects of our study package—particularly the open science practices. Therefore, as you recommended, we have added a new checklist column that indicates whether a given study has utilized open science practices (i.e., preregistrations and open data). The addition of this new column further clarifies how our work goes further than the vast majority of prior research.

Tobias Otterbring

Reviewer #4 (Remarks to the Author):

Thank you for the opportunity to review the manuscript, A Transparency Statement Improves Community-Police Interactions. The authors triangulate across qualitative, field experimental, and web experimental methods to test the claim that having officers transparently state their intentions when contacting the public has salutary effects on these interactions. This work is timely, theoretically rich, empirically rigorous, and practically useful. The authors advance several fields, and this will be an impactful contribution.

As a substitute reviewer, I was tasked with evaluating the revised manuscript in light of the previous reviewer's comments and the authors revisions. To cut to the chase, I believe that this work is of sufficient rigor and novelty for publication, and the authors have responded to the previous reviewer's comments satisfactorily, with one exception. As I describe below, the authors need to provide more clarity about the relationship and novelty of the work vis a vis procedural justice. While I believe this change is important for understanding the impact of the work, I would consider it a minor revision. Here, I address the reviewer's comments -and authors' response- one by one.

1. First, the authors have pulled out a single component (i.e., demonstration of trustworthy motives) from the broader procedural-justice package that has been tested repeatedly in much stronger field trials, after which they acted like it was something new, calling it "transparent communication of benevolent intent," without even seriously (or thoroughly) discussing the other trials.

This critique has two parts: first, the novelty of the contribution vis a vis other research on procedural justice: whether the authors are claiming discovery of a feature of PJ that other interventions have tested. Second, these data are simply restating past demonstrations of procedural justice.

On the second point, I think the authors provide a compelling response detailing how this examination differs from other examinations of procedural justice. The field experiment here is weighted more towards internal versus external validity of the field experiments by Mazerolle and Telep; however, they capture an extremely rare outcome: citizens' perceptions of the actual interaction intervened on. Likewise, procedural justice is often trained as an orientation, with suggested behaviors; this more precise manipulation of one of those behaviors provides a precise test of one of those behaviors. So, the research is not an inferior version of past experiments, nor is it a retread.

On the first point, I agree with the reviewer in principle (if not vehemence). Communicating trustworthy motives is a core feature of procedural justice, which has been the dominant model of police-citizen interactions for the past thirty years. The contribution here is not a retread, but there is a close conceptual relationship: I would characterize it as unpacking how these motives are communicated -or not- in interactions. That is a contribution (as they point out, the operationalizations of PJ are many and often deployed as a package or with maximal officer discretion), but the introduction as written treats PJ as a parallel approach that is incidentally related to their intervention.

The authors respond that transparency statements are parallel, in that their insight came about from qualitative observation. I do not find this convincing. Ride-along observations reveal a lot of things that research has examined: the investigatory nature of many stops, police officers' construals of communities, and so on. It would be fallacious to claim to discover any of these, as it is that the authors have discovered that officers' motives are often opaque to community members.

I don't think the authors actually believe they've discovered transparency, but they have identified an element of communication that is central but unstudied in how transparency is or isn't communicated. I believe the introduction should more clearly identify this contribution and its relation to the voluminous research on procedural justice, earlier rather than later. One simple way to do so is to identify procedural justice as a popular means through which the ideals of trust and legitimacy are pursued, the ambiguities of what PJ looks like in practice, and then their astute observations of how the trustworthy ideals of community policing are lost when officers are not transparent.

We appreciate these thoughtful comments. We agree that clarifying our contribution and how it relates to the existing literature on procedural justice (PJ) is of utmost importance. We very much appreciate your thoughts on how to convey this contribution clearly while not making it sound as though we believe we "discovered" transparency. In fact, your comments have prompted us to further expound upon the grounded theory approach that we took, and how our insights emerged alongside an iterative approach of analyzing our qualitative data and emergent themes and connecting to the extant relevant literature—like that of PJ.

Therefore, we have revised the introduction where we describe both our contribution and how it relates to existing PJ research. For example, on pg. 8, we now state:

> **"Emerging from this qualitative work, one prominent theory justified our hypothesis that a transparency statement could improve community-police relations. This is procedural justice (PJ) theory, which is a prominent framework for explaining the conditions under which people come to view authority figures like police as legitimate. A key tenet of theory on PJ is that how community members are treated during interactions will affect their perceptions of and attitudes towards the police (e.g., trust) over and above the outcomes of the interactions. In particular, a crucial component of the PJ framework is that when police communicate trustworthy motives, community members will view police as legitimate and be more likely to cooperate with police even if the overall outcome of the interaction is negative (e.g. ending in arrest). Thus, PJ theory supports the hypothesis that a transparency statement will build trust between police and their communities. This is because a transparency statement involves an officer making their benevolent intentions clear from the start of an interaction.**
>
> **Although the transparency hypothesis is consistent with the tenets of PJ theory, our research is also a contribution to the literature on PJ theory. This is because the current literature does not provide specific and concrete ways for officers to**

communicate trustworthy motives in their everyday interactions. If research could show that a short transparency statement was effective at increasing trust in community-police interactions, it would be a contribution because it could help to explain why programs inspired by PJ theory in real-world settings have led to mixed results with a high degree of unexplained heterogeneity (I^2 = 95%; see also Table S2 in SI). This high degree of unexplained heterogeneity suggests that research has not yet illuminated which contextual features predict whether PJ treatments will yield their intended effects (e.g., the type of interactions being studied). One reason why PJ programs might be inconsistently effective is because they did not specify, concretely, how officers might best communicate trustworthy motives. Instead, PJ programs have typically involved training officers on all the many different elements of PJ theory while also giving officers discretion in how to implement the recommendations. Thus, a key contribution of our bottom-up approach is to identify the potential impact of the precise language officers might use to communicate trustworthy motives in their interactions with the community."

2. Second, beyond the field trials discussed above, there is a huge body of survey experiments that basically do the same thing (but with better interventions and better outcomes) than the current study does in experiments 2 through 7 (e.g., Johnson et al., 2017; Lowrey et al., 2016; Maguire et al., 2016; Roche, 2019). That is, the last six experiments in the current study look very similar to those that appear in mid-level special journals, and really add nothing new over the findings in those mid-tier articles.

This objection is, as the authors point out, quite vague. What these experiments add is an unpacking of the process believed to underlie the authors' field experiment. The studies the reviewer cites lack the field experimental component of this package. The triangulation of designs is a strength of this paper.

We are grateful for this comment.

3. Third, and this is very important, the key finding from past experimental work—both from the field trials and survey experiments—is that procedural justice interventions, like transparency statements demonstrating trustworthy motives (or benevolent intent), mainly affect encounter-specific attitudes, but have little effect on general policing attitudes (e.g., Johnson et al., 2017; Lowrey et al., 2016; Maguire et al., 2016; Sahin et al., 2017). Nagin and Telep (2016, p. 10) emphasize this point: "There was a significant impact only on these encounter-specific outcomes and not on more general views about the police."

This is a huge problem for the current study where the entire focus is on encounter-specific attitudes and little attention is given to general policing attitudes. That is, in the current study the key outcomes are perceptions of the specific officer in the interaction (e.g., p. 10: "The officer just wanted to make friendly conversation"). The outcomes are NOT general policing attitudes. This is true even in the later experiments where some questions ask about the specific officer or "another officer from this organization" (e.g., how likely to "wave if you see this officer", p. 30). So, what the study basically shows is that if an officer explicitly says they just want to make friendly conservation (e.g., "I'm just out and about walking around, talking to people in the

community"), then civilians are more likely to 7 believe that the specific officer is just trying to have a friendly conversation. Although this is interesting, it is not groundbreaking.

As the authors address, the procedural justice literature is mixed as to whether and when elements of specific encounters generalize to views of organizations and institutions. I also agree with them that the intent behind their study is to understand what features of conversations lead to these impressions of trustworthiness. The focus on interactions is not a fatal flaw for their research question. The linguistic components of interactions which lead to these attributions are not well known and so this constitutes an advance of our knowledge of procedural justice.

Thank you.

4. I would also emphasize that on page 4, the current authors are misrepresenting the literature when they claim that the results of past work are mixed. They cite Bolger and Waters (2019), but that meta-analysis concluded that there was strong support for the theory of procedural justice: "The analysis produced results in support of all three causal chains of Tyler's theory … Citizen perceptions of police procedural justice would appear to have both a direct impact and indirect effect on citizen cooperation, the latter by way of police legitimacy beliefs" (p. 98)…

The authors have addressed this point satisfactorily in their systematic review of the treatment effects of procedural justice.

Thank you.

5. Shockingly, the authors say on pages 15-16 that "people of color and those from lowerSES contexts might have more negative attitudes toward law enforcement" and then they cite a Gallup poll. This is not a "might have" issue. There is a tremendous amount of very strong research that shows quite clearly that such Americans do have more negative policing attitudes and emotions (e.g., Pickett et al., 2022).

The authors address this point, although I would add that the Pickett study they cite (in response to the review comment) in discussing the magnitude of racial gaps in perceptions of the police is slightly misleading. Fear of being killed is a quite an extreme outcome, and other studies have certainly documented large and durable gaps in trust (two examples below). The authors may find them useful in addition to the Pew data.

Hurwitz, J., & Peffley, M. (2005). Explaining the great racial divide: Perceptions of fairness in the US criminal justice system. The journal of politics, 67(3), 762-783.

Peffley, M., & Hurwitz, J. (2010). Justice in America: The separate realities of Blacks and Whites. Cambridge University Press.

Thank you for suggesting these additional relevant citations. We have added them into the revised manuscript in addition to the recent Pew data we had previously referenced (see pg. 3).

6. Finally, I would also emphasize that whereas the authors seemed surprised that they found “no consistent moderation by demographic group” (p. 16), they should not be. Prior research shows that procedural justice has similar effects across demographic groups (Wolfe et al., 2016). And, again, all the current authors are doing is testing the effect of a single dimension of procedural justice, what Mazerole et al. (2013, p. 35, 41) call demonstrating of “trustworthy motives.”

This is another area where there are conflicting results from studies of procedural justice, and the authors’ analysis adds to this discussion and in no way detracts from the argument of the work. I agree with their response.

Thank you!

Reviewer #3 (Remarks to the Author):

The authors have addressed all my former concerns to my satisfaction. As such, my recommendation is to accept the paper in its current form for publication in Nature Communications. Congratulations!

Thank you!

Reviewer #4 (Remarks to the Author):

I appreciate the additional revisions the authors have made to the manuscript. The revised version better situates the work against past research and makes the contributions more legible. I have no further comments and look forward to seeing this work in print!

Thank you!

Reviewer #5 (Remarks to the Author):

1. In the Supplemental Material section Overview of Statistical Approach: BCF Analysis the authors describe the model for the Skin Conductance Levels only. This model appears to include person-level covariates only in the beta term (the "prognostic effect" term) and not in the tau term (the "treatment effect" term). This means that age, race, and gender cannot be inferred to moderate the treatment effects, only time. In this sense, this model is "simpler" than it could be, despite the authors remarking that this model was the "most complex" one in that in was for repeated measurements. Why were the x covariates not used in tau? Moreover, was this the case with the other models as well? I cannot discern because those models are not reproduced. Why not reproduce them all in detail and not just the SCL?

The reviewer is correct: In the prior version of the manuscript, we did not include demographic covariates in the tau term because our key goal was to assess moderation by time in our SCL analyses instead of moderation by person-level covariates. This is because, following from the challenge and threat model of arousal, there are certain timepoints in the interaction in which we were particularly interested (i.e., the initial level of arousal upon the initiation of the interaction by the police officer, and the recovery at the end of the interaction).

We also agree with the reviewer that our wording of "most complex" to describe the repeated-measures model did not accurately capture our meaning and appreciate that the reviewer called this language to our attention. We intended to convey that this model was relatively more complex than the models required to test our single-measurement outcome variables (e.g., self-reported threat and trust).

Therefore, in the revised Overview of Statistical Approach: BCF Analysis section (see pg. 2 in the SI), we now state:

> **"Here we provide more details about how the BCF model was estimated for the skin conductance level (SCL) outcome, because this is the outcome with more complex models since it involves repeated measures in contrast to our other outcomes which involve single measurements. A simpler model that excludes the repeated measures and person-level random effect was fit for the person-level outcomes of threat, trust, natural language, and emotions."**

To the point about the analyses themselves, we did not a priori anticipate that treatment effects would vary by the demographic characteristics of participants (i.e., age, race/ethnicity, gender, native U.S. status, or education level). However, over iterations of the review process, we added in exploratory moderation analyses by demographic characteristics at the requests of previous reviewers particularly for the self-report measures (see also our response to Point #2 below).

However, we agreed that it would be beneficial to add analyses to the supplemental material including person-level covariates (i.e., age, gender, race/ethnicity, native U.S. status, and education) in the tau term for the SCL results (see pg. 35 in the SI).

These analyses reveal that all posterior probabilities of difference are ≤ 0.87, failing to reach our threshold for a "strong" probability of difference:

> **"Mirroring our approach for the focal analyses in the main text, we then identified the conditional average treatment effect (CATE) for each demographic subgroup at the beginning of the interaction (i.e., time = 10) and the recovery/end of the interaction (i.e., time ≥ 80) given that these epochs showed the largest evidence consistent with differential patterns of arousal in the transparency statement compared to the control condition. BCF analyses with demographic characteristics (i.e., age, gender, race/ethnicity, native U.S. status, and education) included as simultaneous moderators did not reveal strong evidence of moderation at the beginning (all pr(ATE≠0) ≤ 0.87) or recovery/end of the interaction (all pr(ATE≠0) ≤ 0.87). Results were consistent when instead conducting BCF analyses to determine whether the pr(CATE for first 10 seconds > CATE for recovery) ≠ 0 (all probabilities ≤ 0.62)."**

Finally, to clarify our BCF approach to all reported analyses, in this revised version we have also added descriptions of each of our BCF models and the terms that are included in each (see pg. 2 & pg. 38 in the SI).

For example, on pg. 38 of the SI, we now state:

> **"In Experiments 2-7, following the approach we implemented in Experiment 1, we**

utilized BCF analyses. Specifically, in our BCF models for the person-level self-report outcomes in each study, we included the vector of person-level covariates (*x*), the treatment indicator, *z* and the person-level outcome variable, *y*."

2. This is especially confusing in light of Table S8, which reports on what, if any, moderating effects there are on the other metrics. This table has outcomes on the columns and moderating variables (demographics, etc) on the rows. As far as I could see, there was no methodological explanation of how this table was produced. I was hoping that it was made using the BCF model, but that would only be possible if those demographic variables had been used, and I suspect they were not. As best I can tell, an ad hoc classical procedure was used to produce S8, though it was not (as far as I can find) described in any way, other than one mention of NHST? This should be fleshed out, as the claim of "no moderation" is an important one.

   In my opinion, it would be nice to use the BCF model in the way it was intended, including x in the tau term and exploring the posterior of tau(x).

The reviewer is right that the version of Table S8 included in the prior submission did utilize null hypothesis testing (NHST) analyses. We added these analyses in a previous round of revision in response to a prior reviewer's comment. As we note in our response to Point #1, we did not a priori anticipate that there would be consistent moderation by demographic characteristics across our self-report measures (we specifically delineated this in our pre-registration of this study: http://tiny.cc/VideoPreReg). However, given the exploratory nature of these moderation analyses, we completely agree that it is more appropriate to conduct them utilizing BCF techniques. Therefore, in our revised Table S8, all results were obtained utilizing BCF models. Corroborating the pattern of results that we obtained when utilizing NHST analyses, overall, we do not see consistent evidence of moderation by demographic characteristics (see our revised Table S8).

In the revised SI, we also briefly summarize the pattern that we see in the few cases where there was a strong posterior probability of difference on pg. 38 of the SI:

> **"In the few cases where BCF analyses did indicate a strong posterior probability of difference, the pattern was consistently such that those who would be expected to have worse attitudes towards the police initially were more positively impacted by the transparency statement compared to those who would be expected to have better initial attitudes (i.e., younger adults; members of underrepresented racial/ethnic minority groups; those with liberal political ideologies; those who indicated they had less favorable baseline attitudes towards the police; and those without prior contact with the police)."**

We appreciate that the BCF analyses provide clarity as to which differences may be more (vs. less) robust than they appeared when using NHST. The manuscript and supplemental analyses are now improved because we have implemented the BCF moderation analyses in the supplemental materials in addition to the main effect BCF results that are our focal analyses in the main text. Thank you!

Version 0, Reviewer 3 attachment:

This is my second review of the current manuscript (I previously reviewed an earlier version in a nother journal). Overall, the paper has been significantly improved, mainly because of certain edits and added empirical evidence. In this sense, I thank the authors for being responsive in the way they addressed most of my formerly raised concerns. That said, I think there are several issues remaining that warrant further attention, and I am still not convinced that the current piece meets the extremely high bar for publication in a journal like *Nature Communications*. Unlike my former review, however, I am willing to opt for a major revision recommendation this time, although I perceive such a potential revision as risky. Below are my remaining main concerns and recommendations to the authors.

While some people believe that "words are violence," I personally think there is an important distinction to be made between physical actions and language, which partially motivated my former comment about the lack of clear behavioral evidence in the field study. That does not mean, as the authors seem to imply in a somewhat passive-aggressive tone in their revision notes, that I do not see the harm caused when someone psychologically abuses another individual. Still, the authors have conducted a study on a college campus, with university students likely constituting a large share of their sample, and with all participants being aware of that they took part in a study. Moreover, the primary outcomes are either self-reported, physiological, or language-based variables. I maintain that this limits the behavioral implications that can be drawn from the study in question and its assumed generalizability to the general population. I was also not convinced by the way the authors addressed this concern, and a related one posed by Reviewer 1 (comment nr. 2), regarding certain labels (i.e., "community members") and issues related to generalizability and representativeness. With respect to the label of "community members," would the authors feel equally confident in referring to university students or online panel members as consumers, patients, or any other generic group simply because virtually all students and online panel members are, or have once been, consumers, patients, or something else? My point is not that the field study is *in*valid; rather, my point is that the authors appear to continue "overselling" the field study, including its sample and focal outcomes. What they have done is impressive, labor-intense, and extremely difficult, but this work does not need to be extrapolated beyond its core content (which might well be impressive enough on most points). I understand the authors' opinion that language can be thought of as a behavioral outcome. But if I were to use the authors' own style of posing rhetoric questions (which they seem to do in the revision notes when they ask whether verbal abuse is farther away from a behavioral outcome than physical abuse), then wouldn't it also be possible to label the hand and finger movements needed to reply to survey items in a questionnaire as a form of behavior, if an equally extreme position would be taken in the opposite direction? Few scholars would agree that hand and finger movements constitute compelling behavioral evidence. I agree that language and interpersonal interactions are closer to behavioral outcomes (Baumeister et al., 2022), but I still think "actions speak louder than words." To me, language is often goal-oriented and hence used either to attain rewards or avoid punishments (except for small talk). In other words, language may well guide certain actions, but I view these actions as real, observable behavior more so than I view the words uttered to approach certain goals or avoid threats.

In their revision notes (p. 6), the authors present a table featuring the local police department 2021 community survey results. They then use the percentages in the table to argue for their position that no meaningful differences exist in the prevalence of certain crime types on-campus versus off-campus. I wonder what the "Other" category represents in this table. If, for example, that category

would include murders and/or homicide, then even a 5 percent difference between on-campus and off-campus sites would have implications for life and death, which seems to go against the authors' claims of no meaningful differences between on-campus and off-campus settings. Relatedly, I do not agree with the authors that there are no available studies demonstrating the importance of location on people's responses in a way that might have influenced the field results. For instance, it has been known and shown for half a century that background authority matters, with about two-thirds of participants in Milgram's (in) famous obedience research complying with giving others "lethal" electric shocks when the data were collected at (the prestigious and authority-signaling) Yale University, whereas compliance rates went well below 50 percent when the same study setup was replicated at a (rundown and substantially less prestigious) commercial building located in the downtown shopping area (Milgram, 1974). Thus, I agree with Reviewer 1 that the setting of the field study is problematic if the authors want to draw generalizable conclusions that go beyond the specific sample included in this study, particularly if such conclusions extend to behavioral implications.

Sticking to the field study, the authors state in their manuscript that it "was a precursor to a future program of field research […] that could test the long-term effects of transparency on public safety" (p. 7). If such long-term oriented work now exists as available data, which could counter many of the raised concerns in this (and the former) round of reviews, I strongly recommend the authors to include the described long-term effects in the current package of studies rather than as a separate submission elsewhere. Doing so would materially strengthen the confidence in the authors' findings, not only from a short-term horizon, and would further underscore the societal relevance of their research.

In their revision notes, the authors also state that they are not aware of any studies showing a discrepancy between the findings obtained from lab experiments (where participants know they take part in a study) and field experiments (where participants typically do not know they take part in a study). While there is a positive correlation between findings from lab and field settings at the aggregate level, it is also true that this correlation is substantially weaker in some disciplines, including disciplines that the authors of the current manuscript rely on. For example, Mitchell (2012) reported that 21 of 80 (26.3%) laboratory effects from social psychology switched sign between research settings (lab vs. field).

Regarding my formerly raised point about timing (in the last round of reviews), the authors seem to refute this issue as a meaningful confound by splitting the control condition into three smaller groups ($n = 10$, $n = 11$, and $n = 109$, respectively) depending on whether officers made a spontaneous transparency statement, an elicited transparency statement, or no transparency statement at all. Specifically, they present three p-values of $p = .05$, $p = .10$, and $p = .40$ as evidence against the idea that the timing and nature of the transparency statement could have influenced the results. With or without a Bonferroni correction, I do not think these results and the corresponding p-values provide compelling evidence for the absence of effects, considering the extremely small cell sizes in two of the three groups coupled with the large absolute mean differences between the groups. If the authors decide to retain this position, then at a minimum they need to supplement the (statistically non-significant) p-values with effect sizes.

When it comes to my former question about interrater reliability for the qualitative work (which Reviewer 1 also seemed to comment on), the authors appear to think that I confused the grounded theory approach with a content analysis. I did not. Moreover, the authors seem to imply that my question does not "rhyme" well with established methods for grounded theory, seemingly inferring either that the reliability of the coders is irrelevant to their work or that research based on grounded theory does not include interrater agreement scores. I firmly believe that reliability is important in any scientific investigation (although not always the most important aspect of academic work), and I also note that it is not uncommon to report interrater reliability coefficients in studies using the grounded theory approach (e.g., Dawes & Larson, 2011; see also Armstrong et al., 1997; Malodia et al., 2021).

In my former review, I also explicitly asked the authors to clarify whether there are any arguments for the idea that aggressive policing would *not* generate higher threat perceptions than the transparency and control conditions. The authors seem to have left this comment largely unaddressed by simply disagreeing (without providing any clear justification for the disagreement). I expect that they address this point in case the Editor opts for the revision decision.

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Version 1, Reviewer 3 attachment:

This is my third review of the manuscript, entitled "A Transparency Statement Improves Community-Police Interactions." I was asked to assess whether my formerly raised concerns had been adequately addressed and whether the issues mentioned by Reviewer 2 from the previous round of reviews had also been satisfactorily handled. In my view, the authors have done a tremendous job, and I am favorable toward publication of the current piece if my remaining issues, summarized below, are carefully addressed with corresponding edits in the manuscript and its supplementary materials.

Regarding the claim, "exploratory analyses revealed that the presence of a spontaneous or elicited transparency statement later in the conversation did not have an effect on our key pre-registered outcome measures of perceived transparency ($p = .10$, $\eta^2 = .04$), threat ($p = .40$, $\eta^2 = .01$), and trust ($p = .05$, $\eta^2 = .04$)" please support it with equivalence testing using Bayesian statistics and/or, if applicable, the frequentist equivalence testing approach using the Two-One-Sided-Test (TOST) framework. Also, two of these three effects resemble the typical effect sizes in psychological science (e.g., Funder & Ozer, 2019; Gignac & Szodorai, 2016; see also Götz et al., 2022), which does not seem to provide compelling evidence for the absence of effects. At least this fact deserves to be acknowledged somewhere. Let the reader decide whether these null findings are interpreted as sufficiently strong evidence for the absence of a given effect.

In my former review, I noted that Mitchell (2012) found 21 of 80 (26.3%) laboratory effects from social psychology to switch sign between research settings (lab vs. field). The authors seem to somewhat explain away this fact despite that they draw on findings that are at least partially based on social psychological literature by simply claiming that they are scholars in management and organizational behavior, with this field conveniently suffering less from results discrepancies between lab and field studies according to Mitchell (2012). While I buy the authors' arguments, I think they can provide some more nuance by noting that portions of the literature they draw on are contextually sensitive (Elbæk et al., 2023; Van Bavel et al., 2016), although they, themselves, find consistent evidence for their theorizing.

Regarding Table S2, I think the authors can do more to show how their own research is unique and differs from most other investigations. I suggest that they add their own paper either in the top or bottom row of the table, while also including more columns such that it becomes conspicuously clear that their own work does much more than what is typically the case in most prior related research. Some suggested columns that can be added are main theory, preregistration, open data, mixed-methods investigation, and physiological measures. Obviously, such a table means that the authors would check all boxes, while most (if not all) previous articles would only check a subset of these boxes (for a similar approach, see Andrews et al., 2014).

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